Dhruv Sagar

Robotics Developer · ROS 2, SLAM and Sensor Fusion

Kitchener, ON, Canada · +1 226 868 6431 · dhruvsagar.pro@gmail.com · dhruvel-portfolio.vercel.app · github.com/dhruvel · linkedin.com/in/dhruvel

SUMMARY

Robotics developer working on ROS 2, SLAM and sensor fusion. At Peribott Dynamics I built the real-time mapping stack for a mobile robot base, fusing an RGB-D camera and a 360° LiDAR with RTAB-Map at 20 Hz, plus the health monitoring and diagnostic tooling that made it reliable across several robot platforms. Honours BSc in Computer Science from Wilfrid Laurier University.

TECHNICAL SKILLS

Robotics: ROS 2 (Jazzy), RTAB-Map, SLAM, visual and LiDAR odometry, ICP registration, TF and frame management, sensor integration and fusion (RGB-D, 2D LiDAR, IMU, wheel encoders), Nav2, RViz2, Gazebo, MuJoCo, occupancy grids, scan matching, frontier exploration, pure pursuit

Embedded & hardware: STM32-based custom hardware, Raspberry Pi, I²C, PWM, GPIO, sensor and actuator integration, motor control, embedded Linux, robot commissioning

Languages: Python, TypeScript, Bash, Java, JavaScript, Solidity; C++

ML & data: NumPy, scikit-learn, PyTorch, Matplotlib

Infrastructure: Linux (Ubuntu, Debian), Docker, Portainer, Ansible, Tailscale, Git, colcon source builds

Web: Next.js, React, TypeScript, MDX, Vercel

Testing: Test plans, exploratory testing, reproducible bug reports, automated sensor validation

AI tooling: Claude Code, Model Context Protocol (MCP), agent skills and subagents, Ollama

EXPERIENCE

Robotics Developer · Peribott Dynamics LLP — 2025 – 2026

- Engineered a schema-driven ROS 2 SLAM pipeline that lets a mobile robot map a room in real time, fusing an ASCAM HP60C RGB-D camera and an LDROBOT D500 LiDAR with RTAB-Map at 20 Hz on a 640×480 camera feed.
- Evaluated SLAM libraries after the vendor-recommended ORB-SLAM2 proved outdated, selected RTAB-Map, and wrote the custom launch configuration for the sensor pair; published it as an open guide (github.com/dhruvel/rtabmap_ros2_ws).
- Built a sensor-fusion and health-monitoring layer: a Python ROS 2 node that watches 5 sensor streams at a 2 s reporting interval and detects a failed sensor in under 5 seconds, with configurable ICP registration (0.3 correspondence ratio, 0.02 m voxel size).
- Standardized SLAM deployment across multiple robot platforms with reusable launch configurations (30-frame short-term memory, 5 m local radius, dual odometry at 0.001 linear/angular variance), Python sensor validators and performance-tuning templates.
- Diagnosed and fixed integration failures (a missing LiDAR-to-camera transform, ROS 2 lifecycle activation, camera-LiDAR timestamp sync, a from-source colcon build of RTAB-Map), and wrote Bash diagnostic scripts and a troubleshooting guide.
- Worked on the robot base itself: motor control, wheel-encoder odometry, IMU integration and on-site commissioning.

Sales Representative · Axio, under Inkeros (representing TELUS) — 2026

- Represented TELUS in door-to-door conversations with prospective customers, listening to their needs, answering questions and explaining plan features so residents could compare services.
- Ran customer credit checks, completed account paperwork, opened accounts with payment arrangements and followed up after installation.

Hotel Operations & Accounting · PND Complex — 2024 – 2025

- Managed day-to-day finances for a small hospitality business (accounts payable and receivable, payroll, budget tracking) in line with local tax and reporting standards, and supported hotel operations, reconciling daily takings and vendor invoices and scheduling staff.

Game & Beta Tester (Remote) · Testify — 2020 – 2022

- Tested pre-release games and apps across builds using structured test plans and exploratory testing, finding functional, UI and performance issues before launch.
- Wrote reproducible bug reports (steps, expected vs actual behaviour, screenshots and logs) and verified fixes in later builds, working fully remote against release deadlines.

PROJECTS

Robotics Playground and In-Browser SLAM Demo 2026
dhruvel-portfolio.vercel.app/playground

- Built an in-browser SLAM simulation in about 1,500 lines of TypeScript on MuJoCo 3.14 (WebAssembly): simulated 180-beam LiDAR and drifting wheel encoders, log-odds occupancy grid, coarse-to-fine correlative scan matching, frontier exploration and pure-pursuit control.
- In headless tests it maps 95–97% of the environment with SLAM error under about 25 cm, at about 0.25 ms per step. Part of a seven-demo playground (cave mapping, service robot, 6-joint arm IK, cart-pole LQR, EKF fusion, A* vs RRT* planning).

Raspberry Pi Homelab 2025
github.com/dhruvel/homelab

- Self-hosted Raspberry Pi cluster with one-command installers (Docker, Portainer, Tailscale, Pi-hole, Samba, Plex), Tailscale remote access with no open ports, and Ansible playbooks that upgrade every node and prepare them for Ceph storage.

Blockchain DNS: Defending Against Cache Poisoning · team of 2 2023
github.com/dhruvel/Blockchain-DNS · Computer Security

- Built a DNS resolver (Python, web3.py) that answers from an Ethereum smart contract (Solidity, Hardhat, Sepolia) instead of upstream nameservers, removing the cache-poisoning attack surface; measured about 167 ms vs 46 ms uncached and under 1 ms cached.

DataDissolve: Secure File Deletion for Android · team of 3 2023
github.com/dhruvel/Data_Dissolve · Android Programming

- Android app (Java, SQLite) that overwrites files with Gutmann (35-pass), DoD-style and Schneier patterns before deletion, with PIN login and English, Hindi and Chinese translations.

Machine Learning Classification · team of 4 2023
github.com/dhruvel/CP322_ML_Classification-Project · Machine Learning

- Implemented KNN and logistic regression from scratch in NumPy; tuned scikit-learn ensemble and SVM text classifiers on IMDB and 20 Newsgroups; trained fully connected and CNN models, including an AlexNet-style network, on CIFAR-10 in PyTorch.

Raspberry Pi Breathalyzer 2023
github.com/dhruvel/Breathalyzer-CP320 · Physical Computing

- Ultrasonic proximity sensing, an MQ-3 alcohol sensor read through a PCF8591 ADC over I²C, and a servo-needle readout, in modular Python.

EDUCATION

Honours Bachelor of Science, Computer Science · Wilfrid Laurier University, Waterloo, ON
Relevant courses: Computer Security, Machine Learning, Android Programming, Physical Computing, Professional Computing Ethics

CERTIFICATIONS

- **Anthropic** (Jul 2026): Claude 101; Introduction to Model Context Protocol; Model Context Protocol: Advanced Topics; Introduction to Agent Skills; Introduction to Subagents
- **PADI** (Apr 2025): Open Water, Advanced Open Water and Enriched Air Diver · **HubSpot Academy**: Email Marketing Software (Sep 2026, valid to Oct 2028)

VOLUNTEERING

Wilfrid Laurier University

- Laurier Foot Patrol; pool guard (CPR certified); group exercise trainer.
- Weekly bouldering wall maintenance and map creation.

WRITING & LANGUAGES

Writing: technical posts on RTAB-Map sensor fusion and the MuJoCo SLAM demo · **Languages:** English, Hindi, French